

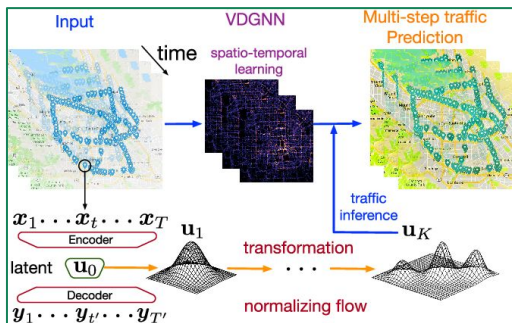
Grounded A.I. in Transportation System

Prof. Jinhua Zhao

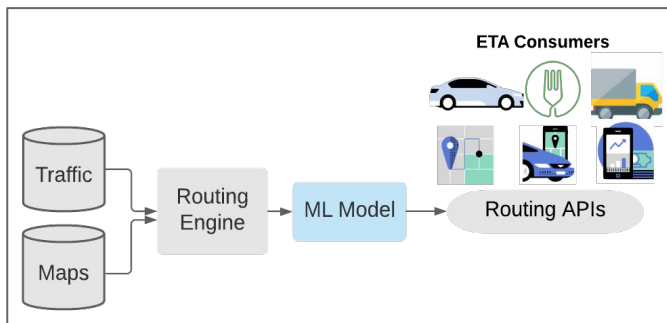
Massachusetts Institute of Technology

Grounded A.I. in Transportation System

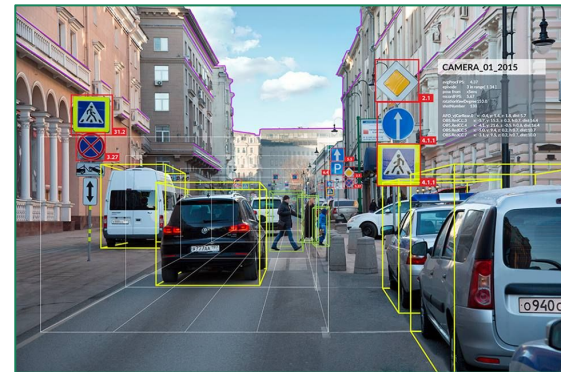
Traffic Management



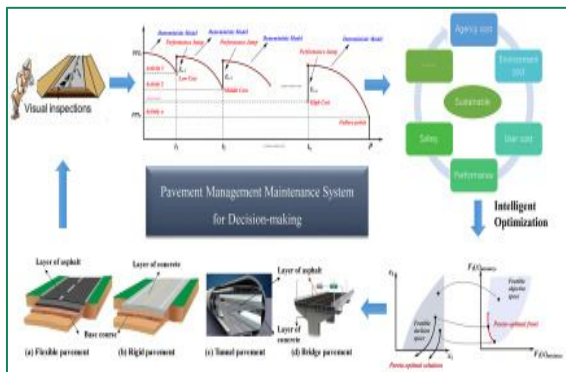
On-Demand Mobility Operation



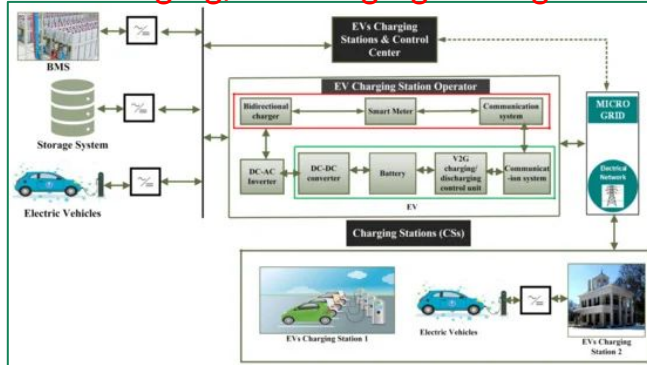
Assisted or Automated Driving



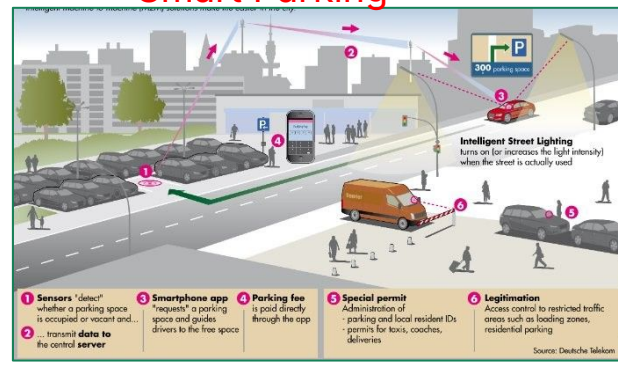
Proactive Infrastructure Maintenance



EV Charging/Discharging Management

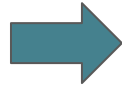


Smart Parking



Fundational Capabilities of A.I.

Data
+
Algorithm
+
Compute

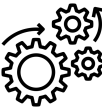


Represent

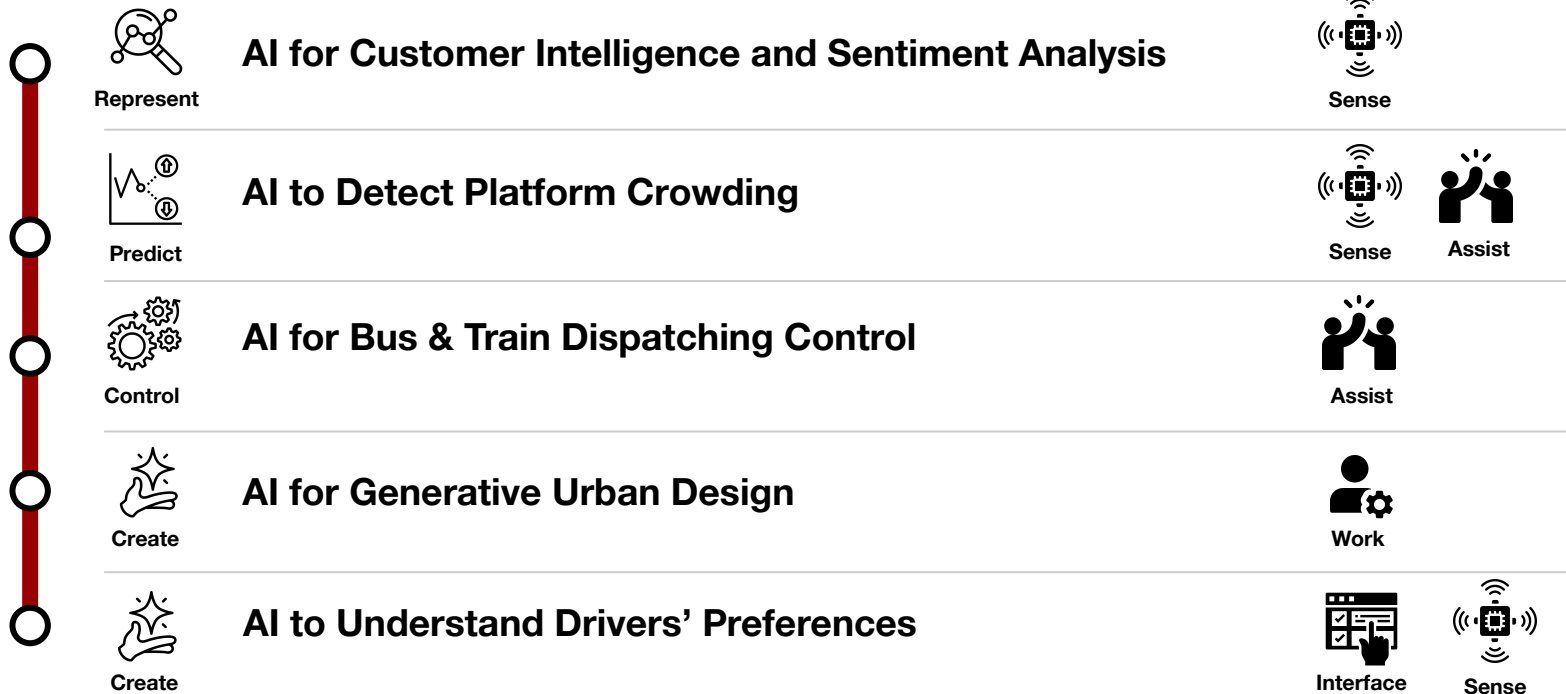
Predict

Control

Create

→ Human-AI Persona ↓ AI Function	 Sense <i>AI used to provide new / live information to existing tasks</i>	 Interface <i>AI used to create easy interactions with humans</i>	 Assist <i>AI used to assist humans and improve performance in tasks</i>	 Instruct <i>AI used to train humans on how to complete tasks</i>	 Work <i>AI used to execute tasks presently done by humans</i>
 Represent <i>AI used to gather, represent, and analyze information</i>					
 Predict <i>AI used to make future predictions based on past data</i>					
 Control <i>AI used to make automatic decisions based on past data</i>					
 Create <i>AI used to create new high-quality content or as general intelligence</i>					

Grounded A.I. in Transportation System



Washington D.C.

Singapore

Chicago

Washington D.C.



Represent



Sense

AI for Customer Sentiment Analysis

What information can you gather from these tweets?



Tony @_tonymm · Apr 12



I'm counting over 25 people waiting for a bus at stop 1003968, [@wmata](#) [@wmataGM](#)! You need to something to improve the S2/S9 route on weekdays for afternoon rush hour! It's insane to spend nearly half an hour to just take a bus!

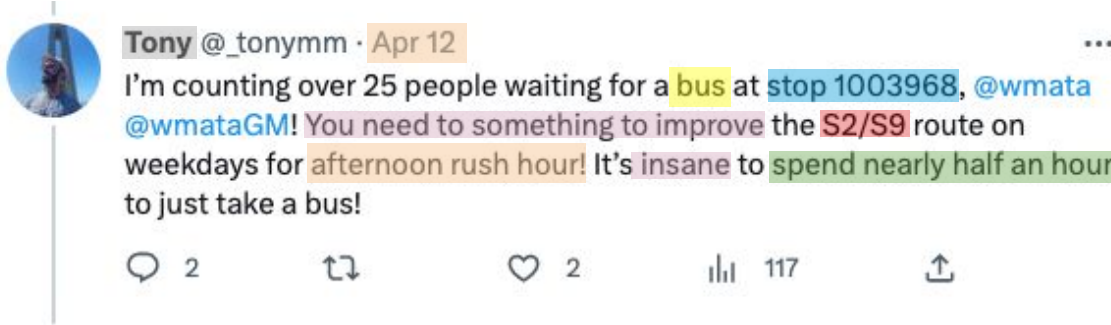


Jess Anders @jess_manders · Jul 16, 2018

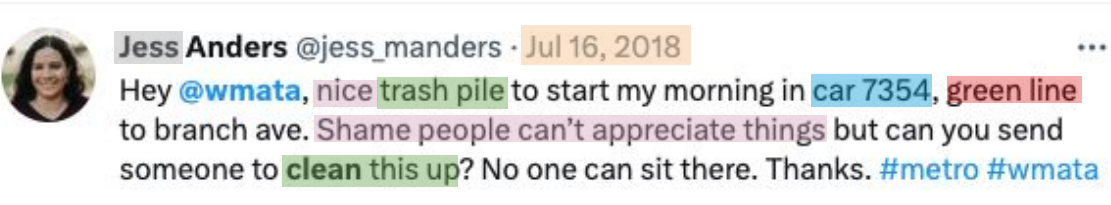


Hey [@wmata](#), nice trash pile to start my morning in car 7354, green line to branch ave. Shame people can't appreciate things but can you send someone to **clean** this up? No one can sit there. Thanks. [#metro](#) [#wmata](#)

What information can you gather from these tweets?



Mode: Bus
Route: S2, S9
Location: Stop 1003968
Time: 4/12/2023 6:48pm
Topic: Schedules / Delays
Sarcasm: Not Present
Sentiment: Negative
Inferred Gender: Male



Mode: Train
Route: Green Line
Vehicle: Car 7354
Time: 7/16/2018 8:03am
Topic: Cleanliness
Sarcasm: Present
Sentiment: Negative
Inferred Gender: Female

Our model **classifies any open-text** feedback with an **accuracy of 94%**

“just want to **shout out the station manager at Dupont Circle for helping me out** when I had some extenuating circumstances. very personable. an asset to the wmata force for sure. I wish I had gotten his name. Thank You!



MetRoBERTa Predicted Topic:
Customer Service

“hey @wmatagm, **does @wmata intentionally slow down trains so customers miss connecting buses or is it a byproduct of their complete indifference to customers?** train sat outside huntington for several mins just long enough so I could see 161 bus pulling away”



MetRoBERTa Predicted Topic:
Schedules/Delays

“hey @wmata, **bus 7108 is out here** on wisconsin in tenleytown just **cutting everyone off and going through red lights**



MetRoBERTa Predicted Topic:
Driving

“one small request @wmata... **8 car trains during rush hour**, on the red line to be exact”



MetRoBERTa Predicted Topic:
Crowding

Our model **classifies any open-text** feedback with an **accuracy of 94%**

“shoutout to @wmata for these fantastic travel information boards at Metro Center. recently had some relatives in town who took the metro from dulles and to see the monuments all weekend, and they were confident first-time riders because of these investments



MetRoBERTa Predicted Topic:
Information Provision

“@wmata it is sweltering hot inside the Archives station. I don’t know how much control you have over this, but a cooler station would make a big difference for customers.



MetRoBERTa Predicted Topic:
Heating/Cooling

“**there is a rat on the south end of the Woodley Park platform.**
Hanging out under the bench at the end of the platform.



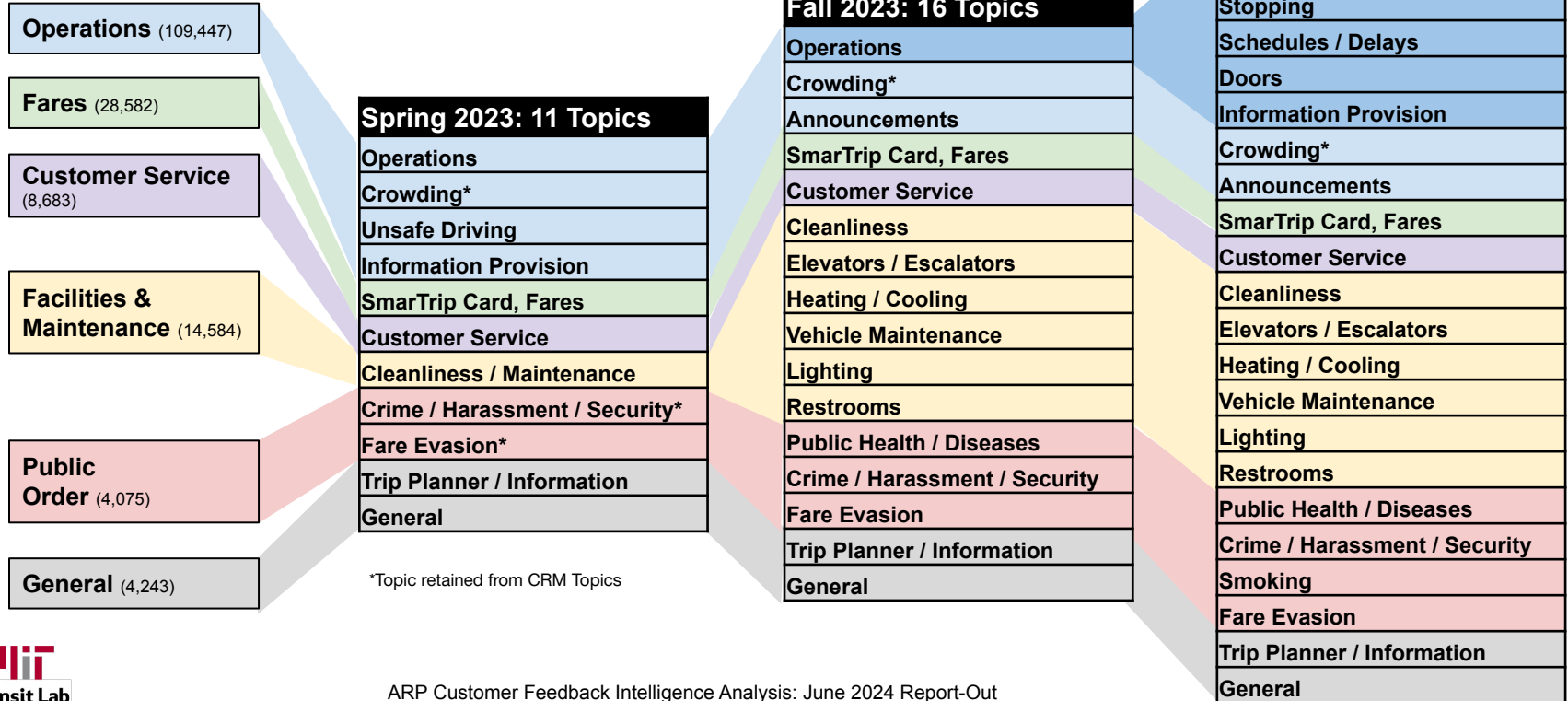
MetRoBERTa Predicted Topic:
Cleanliness

“great smelling elevator that takes me from train platform up to mezzanine to pay and exit the fare gates. thank you!

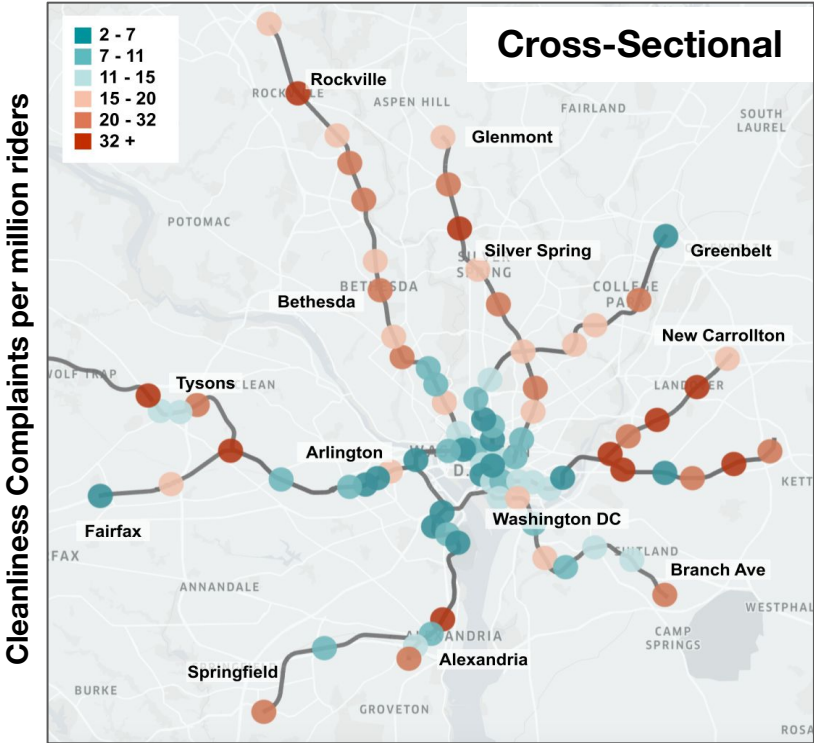
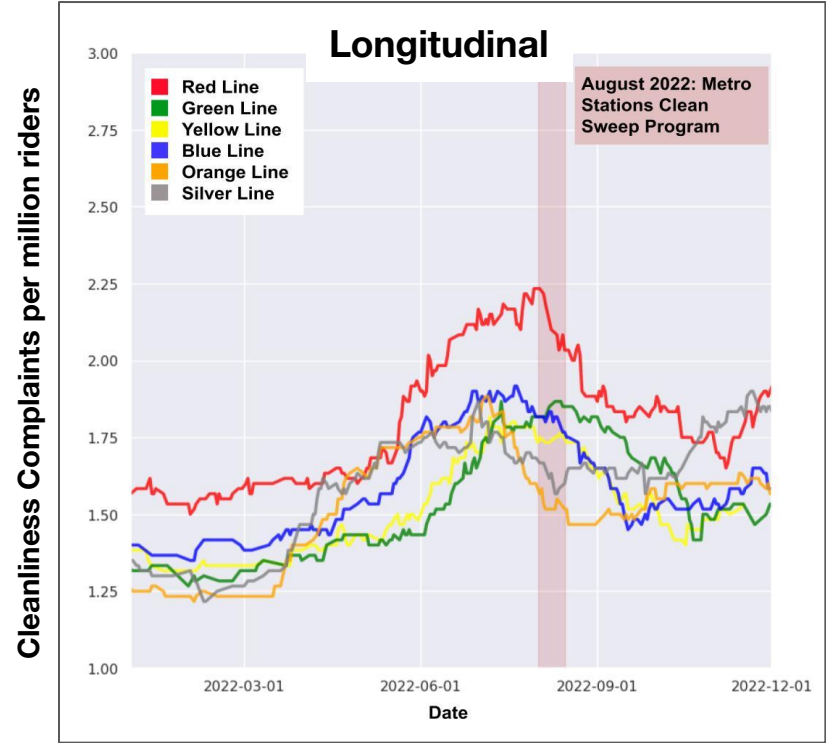


MetRoBERTa Predicted Topic:
Cleanliness

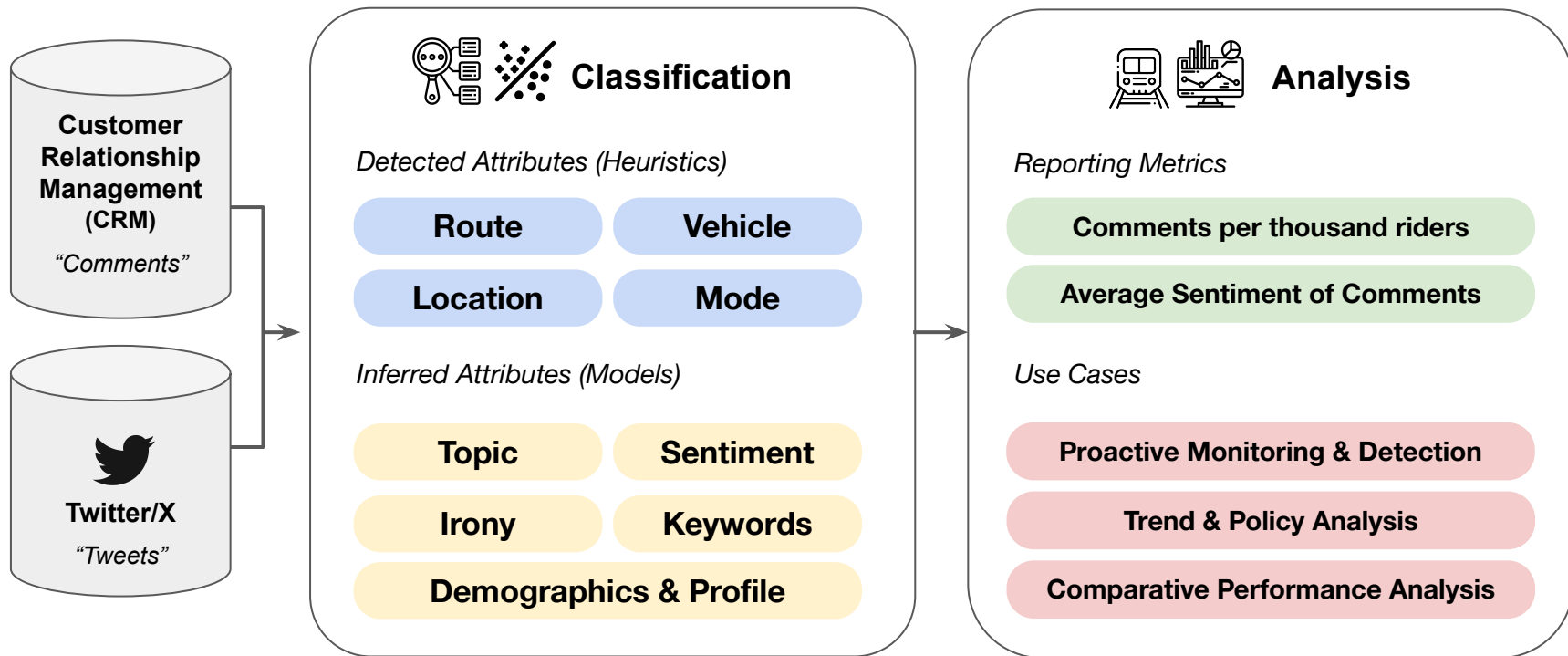
Through further **refinements** and **stakeholder** feedback, we eventually created a **23-topic language model**.



Aggregating & Visualizing Customer Feedback



Unstructured data → Customer insights → Agency actions



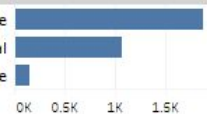
ate Range

months

pics

Service Classification

nts by Sentiment



nts by Keywords

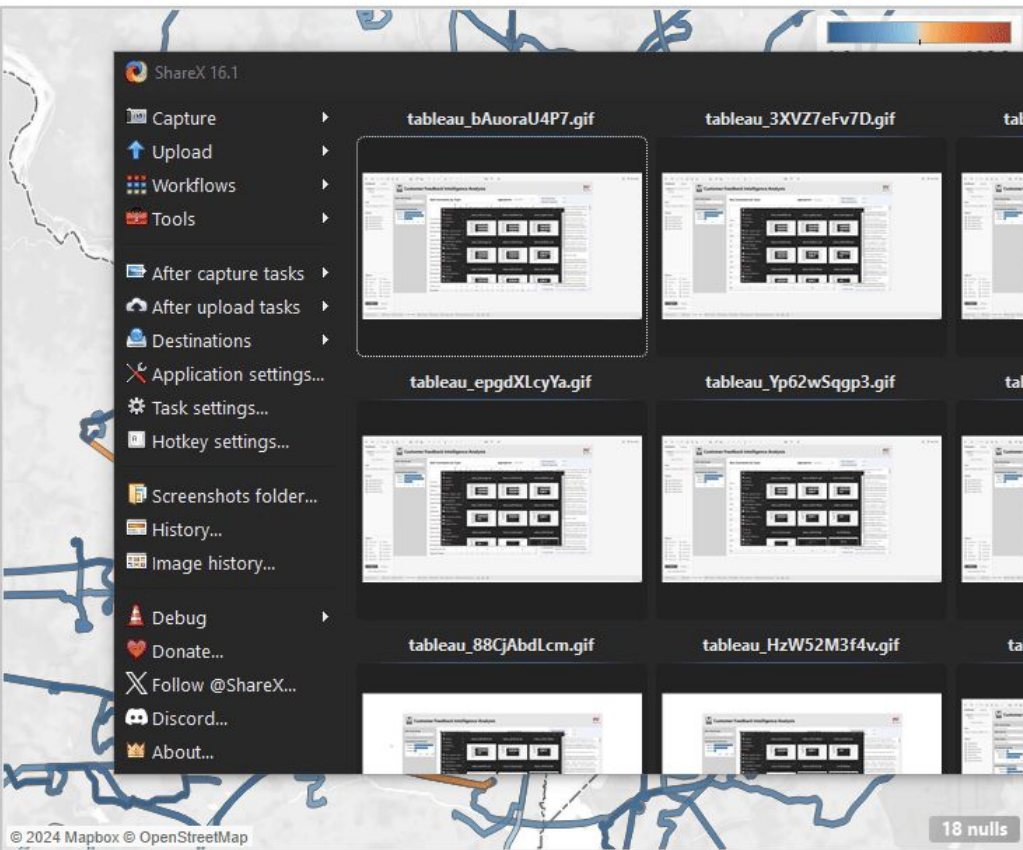
	1,245
	1,042
er	534
o	527
	376
r	369
	314
	273
	263
	262
	255
	224
	220
ers	196
	195
	185

Bus Comments by Route

Color: Total Comments

Total Comments: 3,102

Filtered Comments: 3,102



C-240209-000 why bus stopping so much time on
on drive'

the weekends especially on
e 8:00am h6 bus heading to..

om. learn why this is
caution: this email originate..

2024, the 5:10am f6 to new
did not show up at hyattsvill..

or the r2 bus that leaves at
bus stop # 1003237 at the fo..

05.com.] remove and
ken down bench at wheato..

is very aggressive , rude and
onal, poor service , always l..

rude he doesn't know how
was cursing me he is unprof..

ng the bus scheduled to
stop at 642 am today was a..

assed the stop with no
stop , the bus was not crow..

nk you to the pentagon
upervisor, don, and the 0720..

c705d5308ad1a71 i couldn't
digit number of the bus tha..

e pass me while i was
aiting for the bus

C-240301-000.. i arrived at this bus stop at 7:15 and
expected to take the 7:21 bus to work..

C-240301-000.. i would like to recommend that metro

Singapore

What is NOT borrowable from Singapore?



EPOC EEG
neuroheadset
(128 Hz)



Polar H7 heart rate
sensor (1 Hz)



NeuLog Galvanic
Skin Response
sensor (100 Hz)



Regulate the AV
vs
Upgrade the city

Chicago



Create



Work

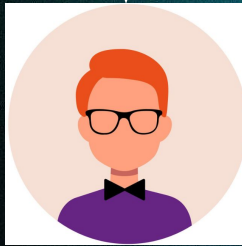
AI for Generative Urban Design

Typical planning process

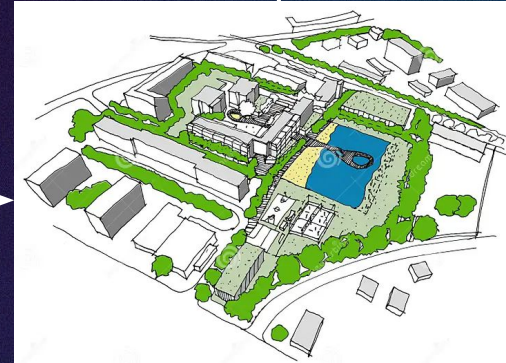
Zoning, Policy, Traffic,
Social interactions,
Natural Environment,
Infrastructure, Public
Engagement,

Feedback and Iterate (weeks and months)

Generate plans, layouts and
visualizations

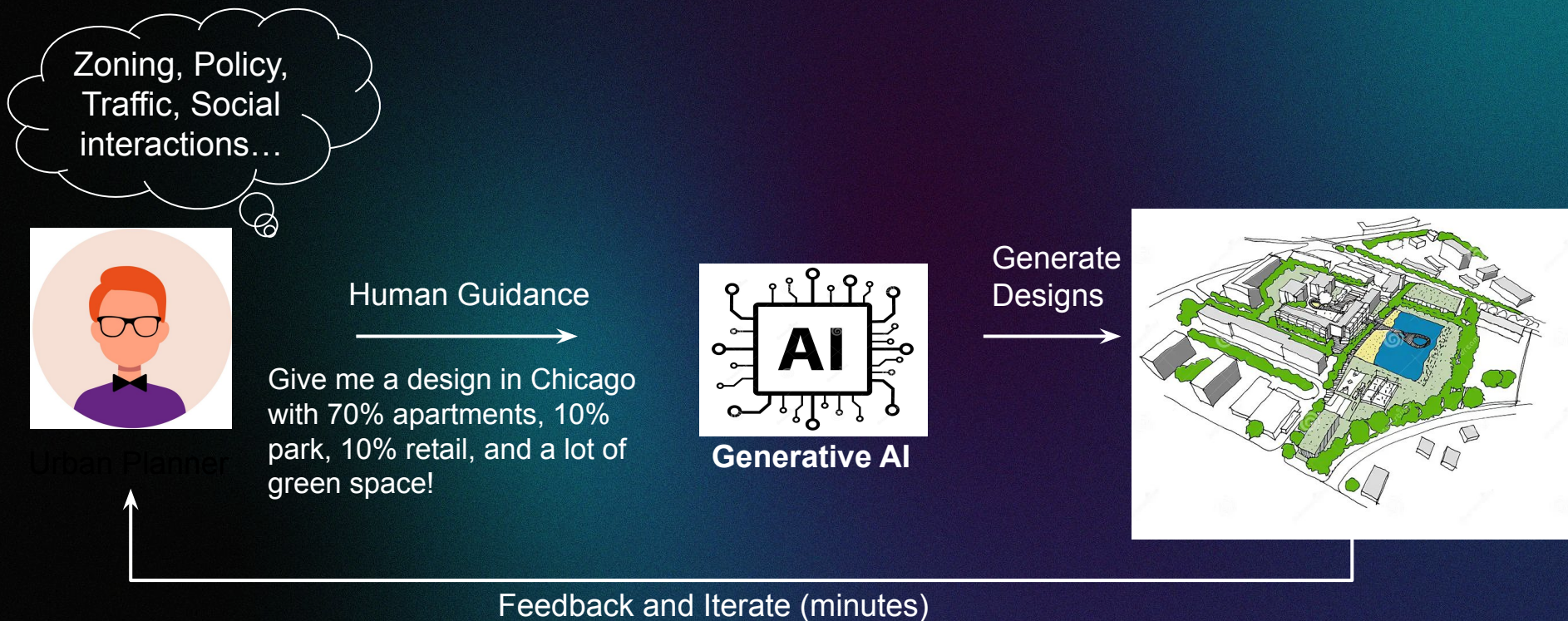


Urban Planner



https://www.123rf.com/photo_16135751_sketch-of-a-new-development-urban-idea.html

The human-machine planning collaboration



A piece of land 35 mile north of downtown Chicago



A target land use

- 35% residential
- 15% commercial
- 10% open parking

What do planners want?

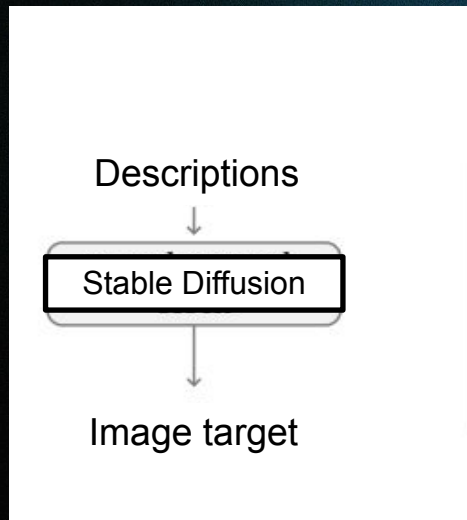
Expectations of Generative Planning Tools

1. Having **control** over AI-generated land-use patterns
2. **Respecting** existing infrastructure and natural environment
3. Learn and apply **different urban textures**

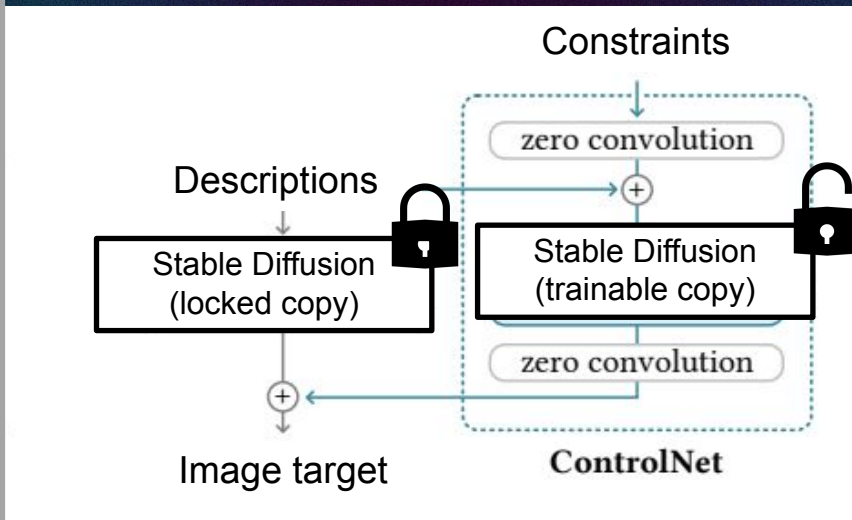
Can A.I. do it?

Fine-tuning stable diffusion using ControlNet

Stable Diffusion



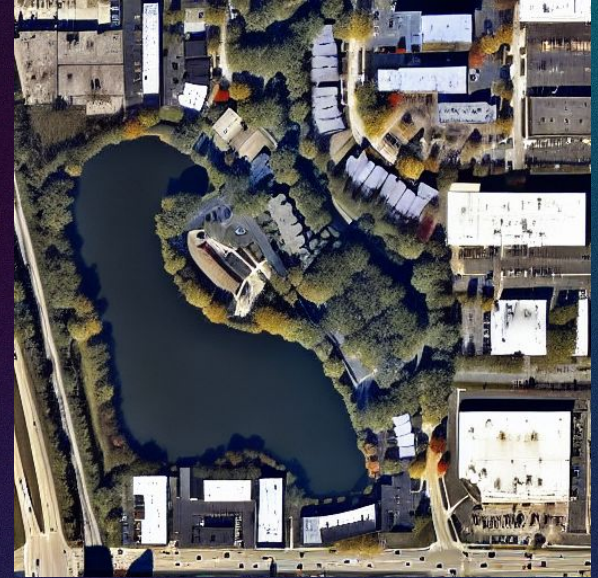
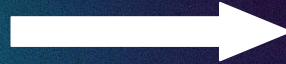
Stable Diffusion with ControlNet



A piece of land 35 mile north of downtown Chicago



- A target land use
- 35% residential
 - 15% commercial
 - 10% open parking



1. Having **control** over AI-generated land-use patterns



35% residential
15% commercial
10% open parking



65% industrial,
20% commercial,
5% open parking



100% residential

2. Respecting **existing** infrastructure and natural environment

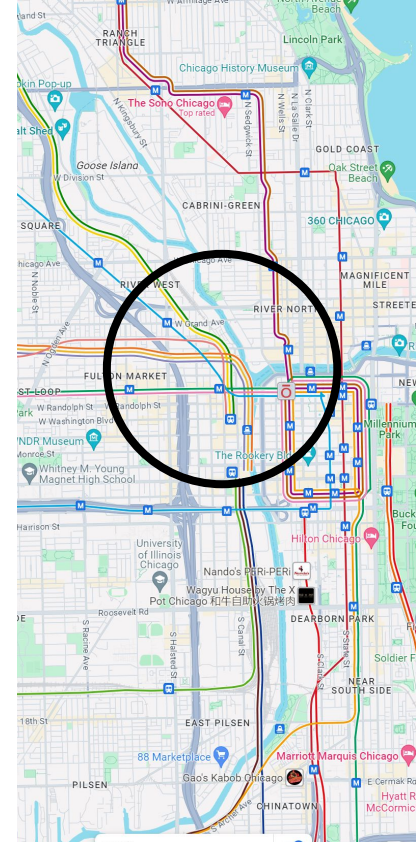
Descriptions

City in Chicago.
35% residential
20% commercial,
5% open parking.
Mix of houses and apartments.
Dense building coverage.

Constraints



Generated Aerial Imagery

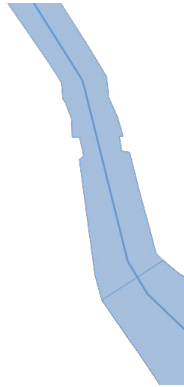


2. **Respecting existing** infrastructure and natural environment

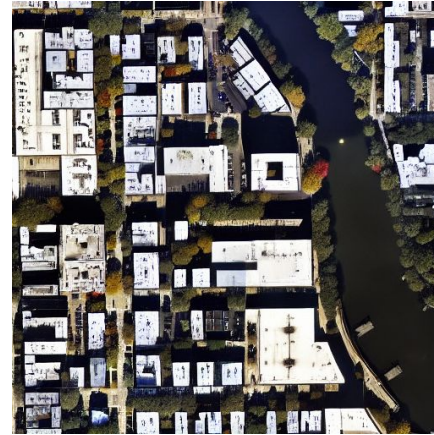
Descriptions

City in Chicago.
35% residential
20% commercial,
5% open parking.
Mix of houses and apartments.
Dense building coverage.

Constraints



Add waterways



Generated Aerial Imagery

2. Respecting existing infrastructure and natural environment

Descriptions

City in Chicago.
35% residential
20% commercial,
5% open parking.
Mix of houses and apartments.
Dense building coverage.

Constraints



Add railway and roads



Generated Aerial Imagery



True Satellite Image

3. Learn and apply **different urban textures**

Descriptions

City in {city name}.

40% residential

15% industrial,

15% commercial,

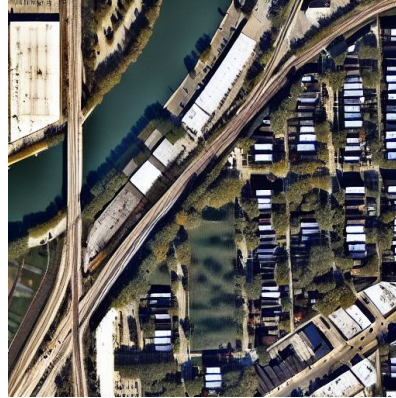
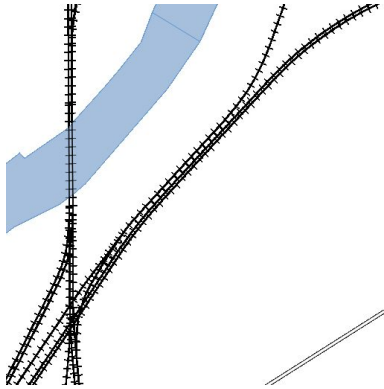
10% park,

5% open parking.

All houses.

Medium building coverage.

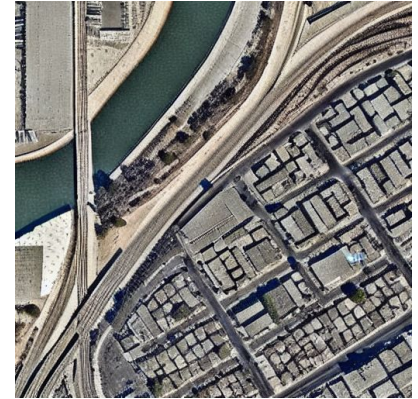
Constraints



Chicago



Dallas



Los Angeles

What do planners want?

Expectations of Generative Planning Tools

1. Having **control** over AI-generated land-use patterns
2. **Respecting** existing infrastructure and natural environment
3. Learn and apply **different urban textures**

What will future planners do?

1. Immediacy changes the dialog

Planners have to be more responsive

2. A.I. enables the public

Planning is a tool to allocate power

3. Essence of planners' job

Design creator → Design critique

Design creator → Design communicator

Design creator → Design negotiator

Customers

Attention —> Intimacy

Employees

Accountability > Productivity

Organizations

Benchmark augmentation vs. automation

Clockspeed

Grounded A.I. in Transportation System

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jinhua@mit.edu

